Complementarity is Abundant, Predictability is Not the Problem: Decomposing Why LLM Routers Gain So Little

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Abstract

LLM routers promise frontier quality at small-model cost. We ask where the promised gain actually goes. We decompose a router's achievable gain over the correct baseline—a *cost-matched, query-independent* policy—into three multiplicative and additive terms: *complementarity* κ (how much oracle headroom the model set offers), *predictability* ρ (what fraction of that headroom is a function of the query at all), and *estimation error* ε (what a fitted router fails to capture). We prove $\rho = 1$ exactly when outcomes are deterministic given the query, give a distribution-free ceiling $\sqrt{\beta(1-\beta)}\,\mathrm{sd}(\delta)$ on every router, and show $\mathrm{sd}(\delta)$ is identified only from *repeated* generations—which no public routing dataset provides. On RouterBench (36,494 prompts $\times$ 11 models, all 55 model pairs $\times$ 8 benchmark families) complementarity is abundant: median $\kappa = 12.11\,\mathrm{pp}$ of oracle headroom at matched cost. Yet the best of four router families captures $0.59\,\mathrm{pp}$, or $\rho_{\mathrm{realised}} = 4.6\%$ of it. We then pre-registered and *refuted* our own explanation. Using $k = 3$ replicate generations of 5,000 prompts across three models—data we generated because no public dataset replicates—we find per-item difficulty is highly *reliable* (0.77–0.98 of outcome variance is stable between-item signal, implying a Bayes-optimal AUC of 0.95–0.99), so $\rho \approx 1$ and unpredictability is *not* the bottleneck. The entire gap is ε: which model will succeed is a stable property of the item that routers cannot read off the prompt. That gap survives eight router families, TF-IDF through a prompted 70B model and a LoRA-fine-tuned 27B model, and a learning curve whose power-law asymptote is 0.74 AUC at 10^9 training items. Routing's binding constraint is generalisation, not noise and not model redundancy.

1 Introduction

A cascade or router sends easy queries to a cheap model and hard ones to an expensive one, then reports savings against always using the expensive model. That comparison is nearly uninformative, because always-cheap and any fixed random mixture also beat always-expensive on cost while looking at nothing. The bar a router must clear is the best *query-independent* policy at the *same expected cost.* Reported gains against always-frontier are not evidence that reading the query helped.

Fixing the baseline is necessary but not sufficient, because it tells you *that* a gain is small without telling you *why.* Three very different worlds produce the same small number. The models might be redundant, so there is nothing to win. The outcome might be a coin flip given the query, so nothing is winnable by any router. Or the information might be there and the router might simply fail to extract it. The first is a model-selection problem, the second a property of the task, the third an engineering problem—and the correct response differs completely.

Contributions.

1. **A decomposition that separates those three worlds** (§2). Achievable gain $= \rho(b)\,\kappa(b)$, and realised gain $= \rho(b)\kappa(b) - \varepsilon$. We prove the ladder $S(b) \le A^\dagger(b) \le A^*(b)$, closed forms for two models, a distribution-free ceiling on *every* router, the exact Bayes-optimal AUC implied by a difficulty distribution, and—the key structural point—that $\rho < 1$ is *exactly* outcome randomness given the query, so an oracle measured on single generations overstates attainable headroom. All propositions are validated numerically against brute-force LP and Monte-Carlo references (14/14 checks, Appendix B).

2. **Complementarity is abundant and routers capture almost none of it** (§3). On RouterBench, median $\kappa = 12.11$ pp $[11.15, 12.92]$ across 55 model pairs $\times$ 8 benchmark families; the best of four router families realises 0.59 pp $[0.44, 0.78]$, i.e. 4.6% [3.6%, 5.9%] of it. The direction is consistent (49/55 pairs positive, 13/55 significant under Holm), so routing does work—it just recovers a twentieth of what is there.
3. **A pre-registered refutation of our own hypothesis** (§4). We predicted low predictability and tested it with $k = 3$ replicate generations of 5,000 prompts $\times$ 3 models. The data says the opposite: difficulty is reliable, $\rho \approx 1$, and our ceiling bound is vacuous at these effect sizes. We report the refutation, and it sharpens the claim rather than weakening it.
4. **The gap is generalisation, and it does not close** (§5). Eight router families—TF-IDF, MiniLM+logistic, MiniLM+MLP, MiniLM+boosted trees, k-NN, random forest, a prompted 70B model, and a LoRA fine-tune of a 27B model—all land in 0.64–0.72 AUC. A learning curve over 250 to 29,000 training items extrapolates to an asymptote of 0.74.

We claim no new router, architecture or algorithm, and we do not claim routing cannot work. κ and ρ are properties of a (workload, model set) pair; the contribution is the instrument that says which term is binding.

2 The decomposition

A query $x \sim \mathcal{D}$; K models; running model m on x yields utility $u_m(x) \in [0, 1]$ (graded correctness) and cost $c_m(x) \geq 0$. Both are random *given* x: decoding is stochastic, and measurably so even at temperature 0 (§4). Write $\eta_m(x) = \mathbb{E}[u_m(x) \mid x]$ and $\gamma_m(x) = \mathbb{E}[c_m(x) \mid x]$. A policy assigns x to a model with probabilities $z_m(x)$, and has value $A(z) = \mathbb{E}[\sum_m z_m(x)u_m(x)]$ and cost $C(z) = \mathbb{E}[\sum_m z_m(x)c_m(x)]$. Three nested classes, distinguished only by what they may look at, with optima at budget b:

$S(b)$ (query-independent), $\quad A^\dagger(b)$ (router: any function of x), $\quad A^*(b)$ (oracle: sees realised u, c).

Proposition 1 (Ladder)**.** *$S(b) \leq A^\dagger(b) \leq A^*(b)$, and all three are concave and nondecreasing in b.*

Definition 1. *$\kappa(b) = A^*(b) - S(b)$* (complementarity) *and $\rho(b) = (A^\dagger(b) - S(b))/\kappa(b)$* (predictability)*, so achievable gain $= \rho(b)\kappa(b)$ and any fitted router $\hat{z}$ realises $A(\hat{z}) - S(b) = \rho(b)\kappa(b) - \varepsilon(\hat{z})$ with $\varepsilon \geq 0$.*

The identity is true by construction; its use is that the three terms are separately measurable and assign blame differently (κ: the model set; ρ: the task; ε: the engineer).

Proposition 2 (What $\rho < 1$ is)**.** *If u_m and c_m are deterministic given x for every m, then $\rho(b) = 1$ for all b. Hence every unit of $\rho < 1$ is outcome randomness given the query.*

Proposition 2 inverts how oracle headroom is normally read. A routing paper's oracle is computed from *one* generation per model per item, so it routes each item to whichever model happened to be right on that draw. When outcomes are noisy, part of what it collects is a coin flip it saw and no router can see. Published oracle headroom is an upper bound inflated by generation noise.

For $K = 2$ (cheap s, dear l, $\beta =$ traffic fraction to l), with $\delta(x) = \eta_l(x) - \eta_s(x)$, $D = u_l - u_s$, and CTE_β the mean of the upper β-tail: $S(b) = \mathbb{E}[u_s] + \beta\mathbb{E}[D]$, $A^\dagger(b) = \mathbb{E}[u_s] + \beta\mathrm{CTE}_\beta(\delta)$, $A^*(b) = \mathbb{E}[u_s] + \beta\mathrm{CTE}_\beta(D)$, so $\rho(\beta) = \big(\mathrm{CTE}_\beta(\delta) - \mathbb{E}\delta\big)/\big(\mathrm{CTE}_\beta(D) - \mathbb{E}D\big)$: predictability is the ratio of tail dispersion of the *conditional mean* to that of the *realisation*. In the binary case the gain of an unconstrained oracle over the better single model is exactly $\min(p_{01}, p_{10})$.

Proposition 3 (Router-free ceiling)**.** *For every router at operating fraction β, $A^\dagger(b) - S(b) \leq \sqrt{\beta(1-\beta)}\,\mathrm{sd}(\delta) \leq \frac{1}{2}\mathrm{sd}(\delta)$, and the bound is attained.*

Proof. For any $z : \mathcal{X} \to [0, 1]$ with $\mathbb{E}z = \beta$, $\mathbb{E}[z\delta] - \beta\mathbb{E}[\delta] = \mathrm{Cov}(z, \delta) \leq \mathrm{sd}(z)\,\mathrm{sd}(\delta)$ by Cauchy–Schwarz, and a $[0, 1]$-valued variable with mean β has variance at most $\beta(1 - \beta)$. Tightness: take δ two-valued and z its indicator. □

Table 1: RouterBench, $\beta = 0.5$: medians over 55 model pairs $\times$ 8 benchmark families (440 cells), with 95% bootstrap intervals over cells. ρ_{realised} is realised gain divided by κ.

Router	out-of-fold AUC	gain (pp)	ρ_{realised}	cells with gain > 0
TF-IDF + logistic	0.7160	$+0.585$ $[0.442, 0.777]$	0.046 $[0.036, 0.059]$	311/440
MiniLM + logistic	0.7081	$+0.455$ $[0.286, 0.609]$	0.034	288/440
MiniLM + boosted trees	0.6772	$+0.402$ $[0.268, 0.544]$	0.033	290/440
MiniLM + MLP (256,64)	0.6626	$+0.226$ $[0.138, 0.326]$	0.017	269/440
Complementarity κ		**12.11** pp $[11.15, 12.92]$		

This is the same covariance object as the cost-accounting correction in §6: routing lives or dies on one covariance, read on two axes.

Proposition 4 (Identification from replicates)**.** *With $k \geq 2$ conditionally i.i.d. replicates per item, binary outcomes, and independence across models,* $\mathrm{Var}(\hat{\eta}_m) = \mathrm{Var}(\eta_m) + \frac{1}{k}\mathbb{E}[\eta_m(1-\eta_m)]$ *and* $\mathbb{E}[\frac{k}{k-1}\hat{\eta}_m(1-\hat{\eta}_m)] = \mathbb{E}[\eta_m(1-\eta_m)]$*, so* $\mathrm{Var}(\delta)$ *is identified by* $\widehat{\mathrm{Var}}(\delta) = \mathrm{Var}(\hat{\eta}_l - \hat{\eta}_s) - \frac{1}{k-1}\overline{[\hat{\eta}_l(1-\hat{\eta}_l) + \hat{\eta}_s(1-\hat{\eta}_s)]}$.

κ is computable from any released outcome matrix. ρ is not: $\mathrm{Var}(D) = \mathrm{Var}(\delta) + \mathbb{E}[\mathrm{Var}(D \mid x)] \geq \mathrm{Var}(\delta)$, so single-generation data bounds $\mathrm{sd}(\delta)$ only loosely, and *every public routing dataset stores one generation per (model, prompt).* The quantity that decides whether routing can work is not recoverable from the data the field publishes. In simulation the uncorrected estimator inflates $\mathrm{Var}(\delta)$ by 2.33× (Appendix B).

Proposition 5 (Bayes-optimal AUC)**.** *With* $a = \mathbb{E}[\eta(X)]$ *and* X, X' *i.i.d.,* $\mathrm{AUC}^{\star} = \frac{1}{2} + \mathbb{E}|\eta(X) - \eta(X')|/(4a(1-a))$*, and* $\mathrm{AUC}^{\star} \leq \frac{1}{2} + \mathrm{sd}(\eta)/(2\sqrt{2}\,a(1-a))$.

Proposition 5 turns a replicate-estimated quantity into a prediction about a number we have measured with many learners—a check that cannot be tuned. Proofs: Appendix A.

3 Complementarity is abundant; routers capture 4.6% of it

Setup. RouterBench [Hu et al., 2024]: 36,494 prompts × 11 models = 401,434 inference outcomes with per-item graded score and per-item dollar cost, grouped into 8 benchmark families with $n \geq 100$ (MMLU 14,042; HellaSwag 10,042; GSM8K 7,450; ARC 1,470; Winogrande 1,267; Chinese 785; other 931; MBPP 427). 21.1% of score cells are fractional partial credit, so we run the decomposition on graded utilities in $[0, 1]$, which needs no threshold. We verified these fractional scores are *not* replicate means—GPT-4 is modal at 0.75 and Mistral-7B at 0.25 with almost no mass at 0 or 1, and only one response is stored per model—so RouterBench cannot support Proposition 4.

We compute $S(b)$ exactly (upper concave envelope of the per-model (cost, utility) points) and $A^*(b)$ exactly (Lagrangian solution of the multiple-choice knapsack LP on *per-item* costs), for all 55 unordered model pairs and the full 11-model problem, over $\beta \in \{0.1, \ldots, 0.9\}$; headline numbers are at $\beta = 0.5$, where Proposition 3's cap is largest and routing is therefore most favoured. Four router families are fit with 5-fold cross-validation grouped by item and stratified by family, so every router trains on 29,195 items. Routers commit using their predictions and per-model *mean* costs (a deployed router cannot know an item's token count in advance) but are charged *realised* per-item cost.

Table 1 is the central empirical result. There is 12.11 pp of oracle headroom over a cost-matched query-independent policy, and the best router takes 0.59 pp of it. The effect is real: 49 of 55 pairs show a positive mean gain, 28 have raw $p < 0.05$, 26 survive Benjamini–Hochberg and 13 survive Holm–Bonferroni at family-wise $\alpha = 0.05$. So this is not a null result about routing; it is a statement about magnitude. Reading gains against always-frontier hides a factor of roughly twenty.

Note also that the MLP is the *worst* of the four despite the most capacity—the same inversion we observe in-house, and the first hint that the binding constraint is not model class.

Table 2: Replicate-based decomposition, $k = 3$. *Reliability* is the share of per-item outcome variance that is stable between-item signal rather than regeneration noise. AUC* is the Proposition 5 point estimate under a Beta fit to the replicate-identified moments.

		Tier 1 (9B)		Tier 2 (20B)		Tier 3 (GPT-4o)
Set	n	reliability	AUC*	reliability	AUC*	reliability
pool, T=0.7	5,000	0.895	0.992	0.873	0.987	0.942
test, T=0.0	364	0.772	0.948	0.902	0.992	0.983

4 Predictability is not the bottleneck: a refuted hypothesis

We pre-registered the hypothesis that ρ is small—that per-item success is weakly predictable in principle—and a falsification rule for it. To test it we generated $k = 3$ replicates for every (item, model) over 5,000 prompts and three models at temperature 0.7 (45,000 calls) and 364 prompts at temperature 0 (3,276 calls), on HumanEval, MBPP, MMLU and GSM8K with execution- or exact-match grading. **The hypothesis was refuted.**

Table 2 says that whether a model answers an item correctly is not a coin flip: 77–98% of the outcome variance is a stable property of the item, and the implied Bayes-optimal AUC is 0.95–0.99. So $\rho \approx 1$: the information a router would need *is* there. Our pre-registered consistency check predicted the replicate-implied AUC* would land within 0.05 of the 0.65–0.68 plateau measured by four learners; it came out 0.99 against 0.68, a miss of 0.31, and is recorded as inconsistent. We included that check because it could embarrass us, and it did.

Correspondingly Proposition 3's ceiling is *vacuous* at these effect sizes: at $\beta = 0.5$ it evaluates to 8.4–15.6 pp against a single-draw κ of 3.3–9.5 pp, a median ratio of 2.82. The inequality is correct and attained on a two-point distribution, but real δ is diffuse, and Cauchy–Schwarz loses exactly that slack. We report this rather than reframing it.

Two things do survive. First, noise inflates the reported oracle: a policy allowed to peek at one graded outcome per model—information no deployed router has—captures only 34–55% of the single-draw κ when evaluated on held-out replicates. Second, 8–35% of the variance of the observed advantage D is generation noise, and the share scales with output length, from 8% for terse GPT-4o to 35% for a long-reasoning 9B model that changes its graded verdict on 19.8% of items between identical temperature-0 calls.

5 The binding term is generalisation

If $\rho \approx 1$ and routers capture 4.6% of κ, then ε is essentially the whole story: difficulty is a stable property of the item that routers cannot read off the prompt. Two experiments ask whether that is fixable.

More data? A learning curve on RouterBench (fixed 20% held-out set, training subsets from 250 to 29,000 items, repeats at small sizes) gives held-out AUC 0.65→0.708 for MiniLM+logistic. A power-law fit $\mathrm{AUC}(n) = A - Bn^{-c}$ gives asymptote $A = 0.741$, with $\mathrm{AUC}(10^6) = 0.726$ and $\mathrm{AUC}(10^9) = 0.738$: roughly 2.5 doublings of data per +0.01 AUC. Realised gain does improve with data (+1.15 pp at 8k to +2.10 pp at 29k, $\rho_{\text{realised}} = 0.105$), so data helps—but the extrapolated ceiling stays far below what capturing κ would need. (Exploratory; not pre-registered.)

A stronger router? Table 3 walks up eight router families on the identical 1,500-item held-out split. Nothing escapes the 0.64–0.72 band. A prompted 70B model with four in-context exemplars reaches 0.710, +0.029 over the best classical learner—real, but below the +0.05 we pre-registered as the threshold at which "you tested a weak router" would become a live objection again.

The random forest is the diagnostic: it reaches training AUC 0.9999—it memorises the training set perfectly—and still generalises *below* logistic regression. Ample capacity, no held-out gain. That is the signature of a target that is not a smooth function of the input representation, not of an inadequate model class.

Table 3: Router-strength ladder, mean held-out per-tier AUC on the same 1,500-item split. Rows above the rule are from earlier stages of this project on identical data.

Router	Representation	mean AUC
k-NN (k=50)	MiniLM	0.6521
Gradient boosting	MiniLM	0.6547
Random forest (train AUC 0.9999)	MiniLM	0.6703
Logistic regression	MiniLM	0.6816
Prompted LLM, zero-shot	Llama-3.3-70B-Instruct	0.6416
Prompted LLM, 4-shot	Llama-3.3-70B-Instruct	0.7105
LoRA fine-tune	Gemma-3-27B-it	*pending*

6 Matched-cost baselines and a biased cost axis

Two measurement corrections underpin the numbers above and matter independently.

The baseline. On our own 364-item frozen test set, a deployed keyword router is beaten on *both* axes simultaneously by the constant policy "send everything to the mid tier": 86.8% vs 92.3% accuracy at 6.87× the measured dollar cost. Its headline saving against always-frontier is 88.6% in modelled joules but 55.0% in measured dollars—the choice of cost axis moves the number by 4× on identical routing decisions. Applying the same audit to RouteLLM [Ong et al., 2025] on its own released GSM8K data, its router *does* beat a cost-matched query-independent mixture, by a mean of +0.57 pp: positive at 8 of 9 operating points (sign test $p = 0.039$) but significant at none individually ($n = 1{,}307$). So the protocol discriminates rather than rejects— and even the passing case has an effect the field's benchmarks cannot resolve point-by-point, while being reported as "up to 85% cost reduction".

The axis. Pricing a routing policy by multiplying traffic fractions by per-model *mean* cost is exact for a query-independent policy but biased for a router, because routers escalate precisely the items that emit more tokens. The exact correction is $R_{\text{true}} = R_{\text{naive}} + \sum_{i \neq j} \operatorname{Cov}\big(\mathbf{1}\{t^*{=}i, \hat{t}{=}j\},\, e_j(x) - e_i(x)\big)$. On our data this term is 1.9× the regret being reported and flips its sign, reconciling to 2×10^{-16}; on RouteLLM's own GSM8K data the sign flip reproduces and naive accounting overstates savings by up to 6.3%. The bias always favours the router and grows with within-model cost dispersion, so it worsens as the field routes among reasoning models with variable-length thinking budgets. All routers in this paper are charged realised per-item cost.

7 Related work

RouteLLM [Ong et al., 2025] and FrugalGPT [Chen et al., 2023] report large savings against always-frontier; we re-audit the former against a cost-matched baseline. RouterBench [Hu et al., 2024] supplies the outcome matrix we decompose and proposes its own routing theory, but prices policies with per-model mean cost and does not separate complementarity from predictability. Concurrent evaluation-focused work reaches a compatible diagnosis from a different direction: Yuan et al. [2025] identify limited task diversity, imbalanced model pools and oversimplified metrics in existing routing benchmarks and build a larger framework (85 models, 33,337 queries), and Huang et al. [2025] likewise argue current router evaluation is inadequate. Our contribution is complementary and orthogonal: not a better benchmark, but a decomposition that attributes a measured shortfall to the model set, the task, or the router—plus the observation that the term which decides the question requires replicate generations that no such benchmark currently provides. The variance machinery is standard (reliability/ICC; CVaR [Rockafellar and Uryasev, 2000]; the mean–variance bound [Bhatia and Davis, 2000]); multiple-comparison control follows Holm [1979] and Benjamini and Hochberg [1995].

8 Limitations

ρ is inferred, not bounded two-sidedly. Our rigorous ceiling is vacuous here; the claim $\rho \approx 1$ rests on the reliability measurements and on a Beta fit to replicate-identified moments (Proposition 5 requires $\mathbb{E}|\eta - \eta'|$, which $k = 3$ does not identify). The Beta fit is a modelling assumption and is labelled as such.

Replicate assumptions. Proposition 4 assumes replicates are conditionally i.i.d. and independent across models. Positive within-item correlation would understate the noise term and hence *overstate* $\mathrm{sd}(\delta)$ and the ceiling—erring in routing's favour.

Scale of the replicate study. The replicate evidence is 48,276 calls over three models, four benchmarks and \$48.09 of real API spend. Three models is not a model ecosystem; the reliability numbers should be re-measured before being assumed elsewhere. The breadth comes from RouterBench, which cannot replicate; the depth from our own data, which is small. Neither alone suffices, and we do not claim the two settings are exchangeable.

Benchmarks are not production traffic. Single-turn, self-contained, auto-gradable academic tasks. Real routing traffic is conversational and mostly not objectively gradable. Every κ and ρ here is conditional on this item distribution.

Energy is modelled, never measured. Where earlier stages report joules they mean token counts times assumed per-tier rates; the headline comparisons in §6 are therefore stated in measured dollars.

Generality. Propositions 1, 2, 4 and 5 hold for general K; Proposition 3 and the closed forms are two-model statements. Our fine-tuned rung is one LoRA configuration on one base model; a larger or differently-trained router could do better than 0.71, and the learning-curve extrapolation is a power-law fit, not a proof.

9 Conclusion

Router gains decompose into complementarity, predictability and estimation error, and the three assign blame to the model set, the task and the router respectively. Measured across 11 models and 8 benchmarks, complementarity is abundant (12.11 pp) and routers capture 4.6% of it. We expected predictability to be the missing factor, pre-registered that hypothesis, and refuted it: per-item difficulty is highly reliable, so the information is there. What is missing is the ability to infer it from prompt text—a gap that eight router families, a 70B prompted model, a fine-tuned 27B model and a 29,000-item learning curve all fail to close. The practical implication is uncomfortable but actionable: for these workloads, effort spent on better routers is likely dominated by effort spent on choosing a more complementary model set, or on obtaining a signal that a prompt does not contain.

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A Proofs

Proposition 1 (ladder). The inequalities follow from $\Pi_0 \subset \Pi_1 \subset \Pi_2$: a constant is a function of x, and a function of x is a function of (x, u, c). For concavity, A and C are linear in z and each class is convex and closed under mixing, so if z^1 is feasible at b_1 and z^2 at b_2 then $tz^1 + (1-t)z^2$ is feasible at $tb_1 + (1-t)b_2$ and attains at least $tA(z^1) + (1-t)A(z^2)$. Monotonicity holds because enlarging b enlarges the feasible set. $\square$

Proposition 2 ($\rho = 1$ iff outcomes are deterministic given x). If u, c are $\sigma(x)$-measurable then $\Pi_1 = \Pi_2$ as policy classes with the same value and cost functionals, so $A^\dagger = A^*$ and $\rho \equiv 1$. $\square$

Two-model closed forms. Assume constant per-model costs $c_s < c_l$ and $\mathbb{E}[u_l] \geq \mathbb{E}[u_s]$, and let β be the traffic fraction sent to l, so $b(\beta) = (1-\beta)c_s + \beta c_l$. (a) A query-independent policy mixes the two points and its value is linear and increasing in β, so the budget binds and $S(b) = \mathbb{E}[u_s] + \beta\mathbb{E}[D]$. (b) Maximising $\mathbb{E}[u_s] + \mathbb{E}[z\delta]$ over $z : \mathcal{X} \to [0,1]$ with $\mathbb{E}[z] \leq \beta$ is a fractional knapsack whose solution sets $z = 1$ on the largest values of δ, giving $\beta\mathrm{CTE}_\beta(\delta)$. (c) Identical, with the oracle ranking by realised D. (d) Binary case: any policy has value at most $\mathbb{E}[\max(u_s, u_l)] = 1 - p_{00}$, attained by the cheapest-correct oracle; since $\mathbb{E}[u_s] = p_{11} + p_{10}$ and $\mathbb{E}[u_l] = p_{11} + p_{01}$ we get $(1 - p_{00}) - \mathbb{E}[u_s] = p_{01}$ and $(1 - p_{00}) - \mathbb{E}[u_l] = p_{10}$, so the gain over the better single model is $\min(p_{01}, p_{10})$. $\square$

Since $\delta = \mathbb{E}[D \mid x]$ and CTE_β is a coherent risk measure, conditioning contracts it, which re-proves $\rho \leq 1$ independently of Proposition 1.

Note that $\min(p_{01}, p_{10})$ is *not* $\kappa(b)$: it measures headroom over the better single model, whereas $\kappa(b)$ measures headroom over the matched-cost *mixture*, which is worse than the better single model whenever $\beta < 1$. A router can gain over a mixture even when $\min(p_{01}, p_{10}) = 0$, by spending a fixed budget where it does the most good. We report both because $\kappa(b)$ is the quantity in the decomposition and $\min(p_{01}, p_{10})$ is the quantity a reader intuits.

Proposition 3 (ceiling). In the main text. The variance step is the fact that a $[0,1]$-valued random variable with mean β has variance at most $\beta(1-\beta)$, attained by the two-point $\{0,1\}$ distribution [Bhatia and Davis, 2000]. The maximum of $\sqrt{\beta(1-\beta)}$ over β is at $\beta = 1/2$.

Proposition 4 (identification). The first identity is the law of total variance with $\mathrm{Var}(\hat{\eta} \mid x) = \eta(1-\eta)/k$. For the second, $k\hat{\eta} \sim \mathrm{Bin}(k, \eta)$ given x, so $\mathbb{E}[\hat{\eta}(1-\hat{\eta}) \mid x] = \frac{k-1}{k}\eta(1-\eta)$ and $\frac{k}{k-1}\hat{\eta}(1-\hat{\eta})$ is unbiased for $\eta(1-\eta)$. Combining across the two models under conditional independence, with $\frac{1}{k} \cdot \frac{k}{k-1} = \frac{1}{k-1}$, gives the stated estimator. $\square$

Proposition 5 (Bayes AUC). Score by η, which is AUC-optimal. With $u = \eta(X)$, $v = \eta(X')$ write $T_> = \mathbb{E}[\mathbf{1}\{u > v\}u(1-v)]$, $T_< = \mathbb{E}[\mathbf{1}\{u < v\}u(1-v)]$, $T_= = \mathbb{E}[\mathbf{1}\{u = v\}u(1-v)]$. Under the half-credit convention the AUC is $(T_> + T_=/2)/(a(1-a))$, and $T_> + T_< + T_= = \mathbb{E}[u(1-v)] = a(1-a)$. By exchangeability $T_< = \mathbb{E}[\mathbf{1}\{u > v\}v(1-u)]$, so $T_> - T_< = \mathbb{E}[\mathbf{1}\{u > v\}(u-v)] = \frac{1}{2}\mathbb{E}|u-v|$. Solving the two equations gives $T_> + T_=/2 = \frac{1}{2}a(1-a) + \frac{1}{4}\mathbb{E}|u-v|$, hence the closed form; the bound follows from $\mathbb{E}|u-v| \leq \sqrt{\mathbb{E}(u-v)^2} = \sqrt{2}\,\mathrm{sd}(\eta)$. Sanity checks: η constant gives $1/2$; $\eta \in \{0,1\}$ equiprobable gives 1. $\square$

B Numerical validation of every proposition

Because the propositions are load-bearing, each is checked against a brute-force reference before any result depends on it (`stage11_13/s11_validate.py`, results in `s11_validate.json`). All 14 checks pass.

Table 4: Validation suite. "LP" references are `scipy.optimize.linprog` with HiGHS; others are Monte Carlo.

Check	Reference	Result
$S(b)$ exact	LP over the mixing simplex, 300 cases	max err 0
$A^*(b)$ exact	multiple-choice knapsack LP, 120 cases	max err 0, no budget overrun
Ladder $S \leq A^\dagger \leq A^*$	Bayes router vs. oracle, n=20k	no violation
$\rho = 1$ iff deterministic	analytic + Bernoulli noise	exact; $\rho \to 0.258$ under noise
Ceiling never violated	exact top-β tail, 4,000 cases	min slack > 0; max tightness 0.856
Ceiling tight	two-point construction	ratio 0.998
Variance components unbiased	known Beta η, 400 cases	rel. bias 0.14%
sd(δ) unbiased	known η pair, 400 cases	rel. bias 0.55%
Naive estimator inflated	same simulation	Var(δ) inflated 2.33×
Bayes AUC closed form	empirical ROC AUC, 200 cases	max err 0.0035
sd bound dominates	empirical AUC*	min slack 0.036
Beta moment match accurate	empirical AUC*	max err 0.0030
$\min(p_{01}, p_{10})$ identity	union minus best single	max err 0
Naive cost axis exact/biased	simulated static vs. router	1.0% vs. 30.3% rel. error

C Pre-registration and deviations

Success criteria, falsification rules, the benchmark-family map, router families, statistics and the cost gate were committed to git *before* the analyses they govern (`stage11_13/prereg_stage11_13.md`); the commit precedes every result artifact and this can be checked against `git log`. This follows the same discipline as two earlier pre-registrations in the project.

Five deviations are recorded in `stage11_13/DEVIATIONS.md`. The two that affect claims:

D1. H3 ("the router-free bound is $\leq 50\%$ of κ") was **refuted**: the bound is $2.82\times\ \kappa$ at median. The pre-registration named this outcome in advance and required us to retreat from the ceiling framing, which §4 does. The replacement claim — difficulty is reliable but not inferable from text — is the one the data supports.

D2. The pre-registration asserted that a policy seeing one replicate $Y^{(1)}$ upper-bounds $A^\dagger$. *This is wrong*: $Y^{(1)}$ is a realised outcome, not a function of x, so the two information sets are incomparable; only their join contains $\sigma(x)$. The error was caught while implementing Stage 12, before any result depended on it. The quantity is still reported, relabelled as the unbiased value of a feasible peeking policy rather than a bound.

D Per-family RouterBench detail

Full per-pair, per-family, per-β output is in `stage11_13/s11_routerbench_pairs_0shot.csv.gz` (440 pair-family cells × 9 operating points, plus the 11-model problem). Summary statistics and all hypothesis verdicts are in `s11_routerbench_0shot.json`. The 5-shot file is analysed identically as a pre-registered replication and reported in `s11_routerbench_5shot.json`.

Multiple-comparison detail at $\beta = 0.5$ for the best router: 49/55 pairs have a positive mean gain, 28 have raw $p < 0.05$, 26 survive Benjamini–Hochberg at FDR 0.05, and 13 survive Holm–Bonferroni at family-wise $\alpha = 0.05$. p-values are two-sided bootstrap over benchmark families with the router and baseline evaluated on identical items.

E Cost accounting for the new experiments

The project's earlier stages spent $48.09 of real API budget. Stage 13 was pre-registered with a $25 gate on additional spend, enforced in code: every call accumulates the provider's own reported usage and the run aborts if the gate is crossed (`stage11_13/s13_spend.json`).

Table 5: Additional spend for the new experiments.

Item	Volume	USD
Prompted router, zero-shot (Llama-3.3-70B)	4,500 calls	0.58
Prompted router, 4-shot (Llama-3.3-70B)	4,500 calls	2.41
LoRA fine-tune (Gemma-3-27B-it, 3 epochs)	10,500 examples	6.71
Fine-tuned router inference	4,500 calls	*pending*
RouterBench analysis, learning curves, theory	—	0.00
Total additional		***pending***

RouterBench, the decomposition, the learning curves and all validation cost nothing: they are analyses of already-released outcomes and of already-paid-for generations. The expensive part of this project is the replicate data, and it is the part no public dataset provides.

F Reproduction

```
pip install -r requirements-eval.txt
python3 stage11_13/s11_validate.py           # 14/14 proposition checks
curl -L -o external_data/routerbench_0shot.pkl \
  https://huggingface.co/datasets/withmartian/routerbench/resolve/main/routerbench_0shot.pkl
python3 stage11_13/s11_routerbench.py 0shot   # decomposition, 55 pairs x 8 families
python3 stage11_13/s12_ceiling.py             # replicate variance components
python3 stage11_13/s12b_learning_curve.py     # learning curves
python3 stage11_13/s13_llm_router.py probe    # cost projection before spending
```

Seeds, package versions, model strings, endpoints and dataset revisions for all stages are listed in `stage7_10/reproducibility_manifest.md` and `stage11_13/`. Reproducibility hazards are flagged rather than glossed: no provider exposes an immutable model revision, so exact regeneration of any API-derived number is guaranteed by nothing under our control. The RouterBench pickle is pinned by SHA-256 in the result JSON.